

SOUTHEAST SHOAL, ON, Canada

WMO: 714660

Lat: 41.8267N

Lon: 82.4632W

Elev: 641

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.6	(c) 10.2	(d) -5.0	(e) 4.3	(f) 8.3	(g) -0.5	(h) 5.5	(i) 13.0	(j) 47.5	(k) 20.9	(l) 43.3	(m) 20.0	(n) 25.8	(o) 270	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	82.2	74.6	80.2	73.3	78.3	71.4	76.8	79.8	74.9	77.5	73.1	76.2	12.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.9	138.7	78.8	74.0	129.9	77.0	72.0	121.1	75.4	40.7	79.8	38.8	77.6	37.1	76.4	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 40.1	(b) 35.7	(c) 31.5	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	27.6	29.7	34.5	44.5	56.8	67.1	72.2	71.9	65.3	55.0	43.0	32.8
	DBStd	17.75	10.01	10.11	8.56	6.73	7.38	6.44	5.19	4.38	6.40	6.84	7.47	8.90
	HDD50	2759	696	569	487	188	18	0	0	0	2	32	231	536
	HDD65	6096	1160	988	947	614	273	53	8	4	73	316	661	999
	CDD50	2809	1	1	6	24	229	512	688	678	461	186	20	3
	CDD65	670	0	0	0	0	19	116	231	217	82	5	0	0
	CDH74	2111	0	0	0	0	30	291	971	704	114	1	0	0
	CDH80	198	0	0	0	0	1	35	114	42	6	0	0	0
Wind	WSAvg	16.2	19.8	16.9	17.0	16.5	14.6	12.8	12.8	12.6	15.0	17.8	19.1	19.5
Precipitation	PrecAvg	37.6	2.5	2.2	2.6	3.6	3.7	3.9	3.7	3.5	3.5	2.8	2.8	2.8
	PrecMax	45.7	5.6	5.3	5.4	5.4	7.2	6.3	8.6	8.5	8.1	4.7	5.5	7.2
	PrecMin	26.6	0.7	0.6	1.1	2.0	1.2	1.3	1.6	0.8	1.2	0.7	0.8	1.2
	PrecStd	4.9	1.2	1.2	1.1	1.0	1.7	1.4	1.6	1.8	1.6	0.9	1.0	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.3	56.2	61.3	65.6	77.4	83.9	85.9	83.2	79.9	71.8	61.1	56.0
		MCWB	51.4	48.2	50.9	55.0	65.5	72.9	77.0	74.5	71.3	64.4	54.4	54.4
	2%	DB	48.1	50.3	55.2	60.7	73.8	80.1	82.7	81.1	76.7	69.0	58.1	51.7
		MCWB	46.3	47.6	48.9	52.3	64.0	71.4	76.1	73.3	69.0	62.5	52.2	48.9
	5%	DB	44.6	46.4	49.6	56.9	70.8	77.1	80.9	79.4	75.2	66.3	55.6	47.4
		MCWB	43.2	43.5	44.2	49.9	62.4	70.6	75.4	72.1	68.5	60.5	50.3	44.0
	10%	DB	39.9	41.6	45.1	53.6	67.2	74.8	78.8	77.8	73.6	63.7	53.2	43.9
		MCWB	37.8	37.8	40.3	47.6	60.6	69.3	72.6	70.9	66.9	57.7	48.5	40.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.9	49.3	54.1	57.5	68.3	76.1	79.6	78.8	74.5	66.9	57.0	54.1
		MCDB	52.4	52.5	59.2	62.5	73.2	78.8	82.5	80.6	75.8	69.2	59.9	55.4
	2%	WB	45.8	43.2	50.0	54.0	65.7	73.7	77.4	76.4	72.3	64.3	53.9	47.6
		MCDB	47.0	45.8	54.6	57.2	72.0	76.9	80.7	78.7	74.5	67.7	56.9	49.2
	5%	WB	41.3	38.5	45.4	50.9	63.2	71.8	75.6	74.5	70.2	61.3	50.9	42.3
		MCDB	42.3	40.2	49.8	55.2	68.5	75.2	78.6	77.2	73.4	64.9	55.0	44.3
	10%	WB	35.9	36.2	41.3	48.3	60.9	70.0	73.4	72.6	68.4	58.8	48.5	39.3
		MCDB	37.4	37.8	45.2	53.1	66.4	73.6	76.6	76.2	72.4	62.6	52.4	41.8
Mean Daily Temperature Range	5% DB	MDBR	8.8	8.1	9.1	9.5	9.1	7.8	7.1	6.8	7.4	7.9	7.4	7.0
		MCDBR	12.3	11.1	16.8	15.4	13.2	10.0	8.7	7.6	7.7	9.7	10.3	9.0
		MCWBR	13.6	13.4	12.8	10.4	8.1	7.2	7.9	7.8	7.5	8.4	9.6	9.5
	5% WB	MCDBR	14.0	11.2	17.1	14.6	12.2	9.0	8.2	7.4	7.0	10.3	10.6	9.0
		MCWBR	14.0	11.2	14.1	11.5	9.0	7.8	8.6	8.7	8.5	9.7	10.1	9.6
Clear-Sky Solar Irradiance	taub	0.307	0.332	0.356	0.397	0.430	0.444	0.438	0.424	0.382	0.366	0.325	0.314	
	taud	2.391	2.288	2.303	2.267	2.218	2.228	2.266	2.293	2.388	2.407	2.486	2.445	
	Ebn at Noon	266	274	281	277	269	265	265	266	270	260	258	253	
	Edh at Noon	26	33	37	41	44	44	42	40	34	30	23	23	
All-Sky Solar Radiation	RadAvg	439	702	1086	1458	1724	1929	1933	1692	1360	882	549	375	
	RadStd	48	89	100	116	144	125	106	67	118	70	78	37	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+171	N/A

Nomenclature: See separate page